INCH-POUND
A-A-60035C
12 May 2026
SUPERSEDING
A-A-60035B
17 April 2024

COMMERCIAL ITEM DESCRIPTION

BOLTS, CADMIUM AND HEXAVALENT CHROMIUM FREE

The General Services Administration has authorized the use of this commercial item description by all federal agencies.

1 SCOPE. This commercial item description (CID) covers bolts containing no cadmium or hexavalent chromium in their finishes. This CID was developed to address steel fastener coatings without hazardous substances for use on military ground vehicles.

1.1 Intended use. Bolts contained herein are intended for ground vehicle applications. Bolts described in this CID shall not be used for aerospace applications without design authority approval.

2 CLASSIFICATION

2.1 Types.

Type I	-	DELETED.
Type II	-	NAS1297 - Bolt, Shoulder, Hexagon Head
Type III	-	NAS6603 THRU NAS6620 - Bolt, Tension, Hex Head, Close Tolerance, Alloy Steel, Long Thread, Reduced Major Dia., Self-Locking and Nonlocking, 160 KSI Ftu
Type IV	-	NASM3 THRU NASM20 - Bolt, Machine, Aircraft
Type V	-	NASM21098 - Bolt, Self-Locking, Steel, 160 KSI Ftu, 250 °F, 12 Point, External Wrenching (Externally Wrenching Cap Screws)
Type VI	-	NASM21250 - Bolt, Tension, Steel, External Wrenching, Flanged, 12-Point, 180 KSI Ftu, 450 °F
Type VII	-	DELETED.
Type VIII	-	NASM21134 - Bolt, Tension, Steel, 180 KSI Ftu, 450 °F External Wrenching, Spline Drive, Flanged
Type IX	-	NASM20073 - Bolt, Machine, Aircraft, Drilled Head, Fine Thread

3 SALIENT CHARACTERISTICS

3.1 Materials. Unless otherwise specified herein, the bolts shall be manufactured as specified in the Aerospace Industries Association (AIA) documents listed above by type with the exception of, the plating/finishes shall be free of cadmium and hexavalent chromium finishes as specified herein. The use of recovered materials made in compliance with regulatory requirements is acceptable provided all requirements of this CID are met.

Beneficial comments, recommendations, additions, deletions, clarifications, and any other data that may improve this document should be sent to: U.S. Army DEVCOM Ground Vehicle Systems Center, ATTN: Standardization, FCDD-GVS-SAT, MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or emailed to usarmy.detroit.devcom-gvsc.mbx.standardization@army.mil. Since contact information can change, you may want to verify the currency of the address information using the ASSIST Online database at https://quicksearch.dla.mil.

FSC 5306

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3.2 Dimensions. Dimensions for all designated type of bolts apply after plating/finishing and prior to any application of lubricants (if any). Refer to each individual NAS/NASM specification listed in Section 2.1 for detailed part geometry and tolerance requirements.

3.3 Hexavalent chromium free Part or Identifying Number (PIN) codes and finish/plating options. PIN finish/plating codes for all designated types of bolts:

QX	=	Carbon/alloy steel shall be Zinc Nickel plated, Type K16 in accordance with MIL-PRF-32647.

3.4 Lubrication. Lubrication types (if any) for each type of bolt shall be specified in the referenced AIA/NASM specification (see 2.1).

3.5 Interchangeability. When AIA/NASM specifications list interchangeability parts, do not interchange parts containing cadmium or hexavalent chromium with the cadmium and hexavalent chromium free parts listed herein without design authority approval.

4 REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered, recycled, bio based, and environmentally preferable materials to the maximum extent practicable, in accordance with paragraph 23.403, 23.404(b) of the Federal Acquisition Regulation (FAR).

5 PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Responsibility for inspection. The contractor is responsible for the performance of all inspections (examinations and tests).

5.3 Contractor certification. The contractor shall certify and maintain substantiating evidence the product offered meets the salient characteristics of this CID and conforms to the producer's own drawings, specifications, workmanship standards, and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to perform or witness any of the inspections it deems necessary to ensure supplies and services conform to prescribed requirements.

6 PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

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7 NOTES

The following part or identification numbering (PIN) procedure is for government purposes and does not constitute a requirement for the contractor.

7.1 Tables. The following finish/plating PIN's are cadmium and hexavalent chromium free for designated bolt types.

TABLE I. DELETED

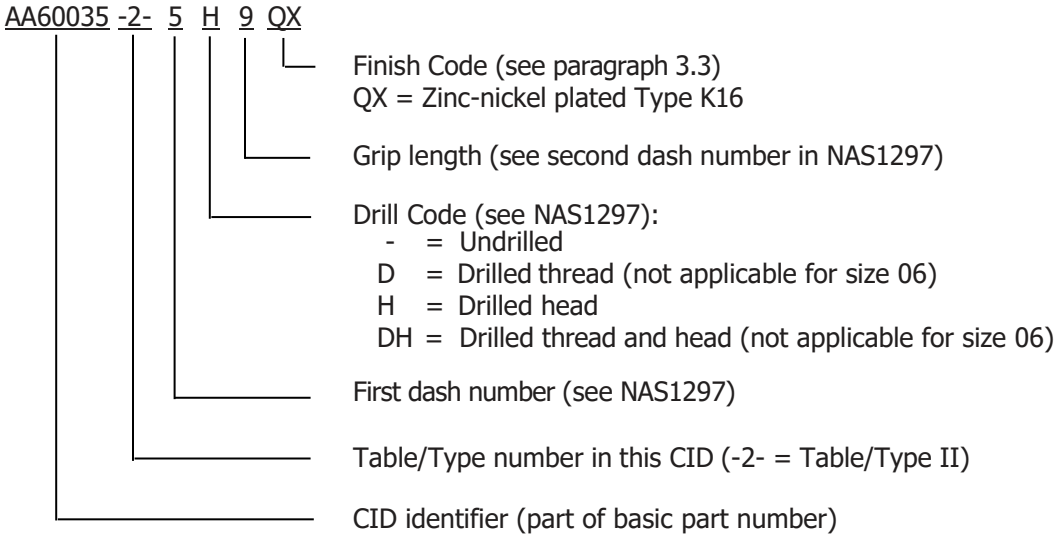
TABLE II. Type II (-2), bolt, shoulder, hexagon head.

(Use NAS1297 for material, heat treatment (if any), and sizes/lengths.)

A-A-60035 Part Number Zinc-Nickel Type K16 Plated (hex chrome-free)	NAS1297 Reference Dash Number Cadmium Plated (see paragraph 7.3)
Alloy Steel	
AA60035-2-06*#QX	NAS1297-06*#
AA60035-2-3*#QX	NAS1297-3*#
AA60035-2-4*#QX	NAS1297-4*#
AA60035-2-5*#QX	NAS1297-5*#
AA60035-2-6*#QX	NAS1297-6*#
AA60035-2-7*#QX	NAS1297-7*#
AA60035-2-8*#QX	NAS1297-8*#
AA60035-2-10*#QX	NAS1297-10*#

* indicates drill code (four potential possibilities)
\# indicates grip length (refer to NAS1297)

PIN construction:



Examples of clean finish PIN's:

AA60035-2-06-4QX = Bolt, shoulder, hexagon head, alloy steel, .1380-32 UNJC-3A thread, undrilled, .260 grip length, zinc-nickel Type K16 plated.

AA60035-2-5D8QX = Bolt, shoulder, hexagon head, alloy steel, .3125-24 UNJF-3A, drilled thread, .510 grip length, zinc-nickel Type K16 plated.

AA60035-2-5H9QX = Bolt, shoulder, hexagon head, alloy steel, .3125-24 UNJF-3A, drilled head, .572 grip length, zinc-nickel Type K16 plated.

AA60035-2-10DH10QX = Bolt, shoulder, hexagon head, alloy steel, .6250-18 UNJF-3A, drilled thread and head, .635 grip length, zinc-nickel Type K16 plated.

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TABLE III. Type III (-3), bolt, tension, hex head, close tolerance, alloy steel, long thread, reduced major dia., self-locking and nonlocking, 160 KSI Ftu.

(Use NAS6603 THRU NAS6620 for material, heat treatment (if any), and sizes/lengths.)

A-A-60035 Part Number Zinc-Nickel Type K16 Plated (hex chrome-free)	NAS6603 THRU NAS6620 Reference Dash Number Cadmium Plated (see paragraph 7.3)
AA60035-3-03*QX	NAS6603*
AA60035-3-04*QX	NAS6604*
AA60035-3-05*QX	NAS6605*
AA60035-3-06*QX	NAS6606*
AA60035-3-07*QX	NAS6607*
AA60035-3-08*QX	NAS6608*
AA60035-3-09*QX	NAS6609*
AA60035-3-10*QX	NAS6610*
AA60035-3-12*QX	NAS6612*
AA60035-3-14*QX	NAS6614*
AA60035-3-16*QX	NAS6616*
AA60035-3-18*QX	NAS6618*
AA60035-3-20*QX	NAS6620*

* represents a placeholder for the configuration options for the bolt, including: drill option, self-locking, grip dash number, and oversize code

PIN construction:

AA60035 -3- 08 H L 22 X QX

Finish Code (see paragraph 3.3)
QX = Zinc-nickel plated Type K16

Oversize Code:
No code = No oversize
X = .0156 Overside Shank
Y = .0312 Oversize Shank

Grip dash number in .0625 increments (see NAS6603 THRU NAS6620)

Feature Code(s):
"-" = Undrilled / No self-locking element
D = Drilled Threads only
DH = Drilled Head / Drilled Threads
H = Drilled Head only
HL = Drilled Head / optional type self-locking element
HP = Drilled head / patch type self-locking element
L = Optional type self-locking element only
P = Patch type self-locking element only

Basic Part Number Code, 03 thru 20 correlating with NAS6603 THRU NAS6620

Table/Type number in this CID (-3- = Table/Type III)

CID identifier (part of basic part number)

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Examples of clean finish PIN's:

AA60035-3-04-10QX = NAS6604 Bolt, .2500-28 UNJF-3A thread, .625 grip, undrilled / nonlocking, zinc-nickel Type K16 plated.
AA60035-3-08H22QX = NAS6608 Bolt, .5000-20 UNJF-3A thread, 1.375 grip, drilled head, undrilled thread, nonlocking, zinc-nickel Type K16 plated.
AA60035-3-08HL22QX = NAS6608 Bolt, .5000-20 UNJF-3A thread, 1.375 grip, drilled head, undrilled thread, self-locking (patch type), zinc-nickel Type K16 plated.
AA60035-3-08DH22QX = NAS6608 Bolt, .5000-20 UNJF-3A thread, 1.375 grip, drilled head, drilled thread, nonlocking, zinc-nickel Type K16 plated.

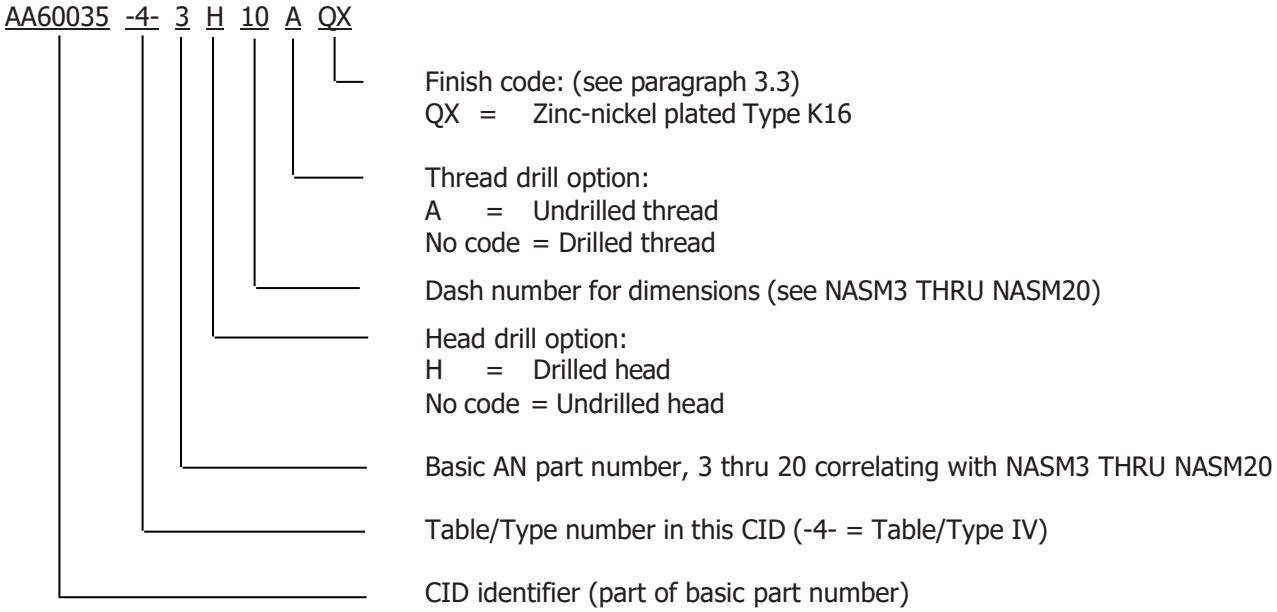
TABLE IV. Type IV (-4) - bolt, machine, aircraft.

(Use NASM3 THRU NASM20 for material, heat treatment (if any), and sizes/lengths.)

A-A-60035 Part Number Zinc-Nickel Type K16 Plated (hex chrome-free)	NASM3 THRU NASM20 Reference Dash Number Cadmium Plated (see paragraph 7.3)
Non-Corrosion Resistant Steel	
AA60035-4-3*	AN3-#
AA60035-4-4*	AN4-#
AA60035-4-5*	AN5-#
AA60035-4-6*	AN6-#
AA60035-4-7*	AN7-#
AA60035-4-8*	AN8-#
AA60035-4-9*	AN9-#
AA60035-4-10*	AN10-#
AA60035-4-12*	AN12-#
AA60035-4-14*	AN14-#
AA60035-4-16*	AN16-#
AA60035-4-17*	AN17-#
AA60035-4-18*	AN18-#
AA60035-4-20*	AN20-#

* See PIN construction for part number breakdown
See NASM3 THRU NASM20 for part number breakdown
(AA60035-4-16 is inactive for new design after June 15, 1966)

PIN construction:



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Examples of clean finish PIN's:

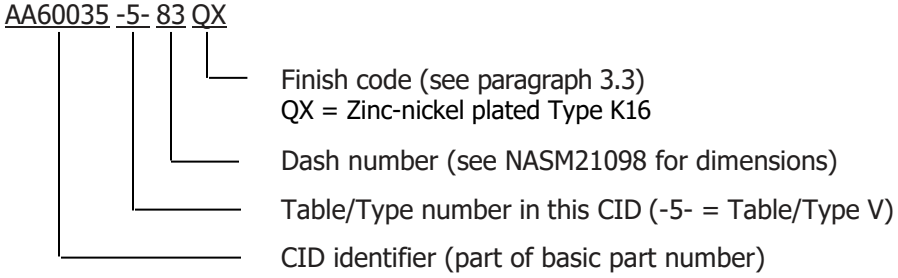
AA60035-4-610QX = NASM3 THRU 20 AN6 non-corrosion resistant steel bolt, .3750-24 UNF-3A thread, 1.078 long, .438 grip, drilled threads, undrilled head, zinc-nickel Type K16 plated.

TABLE V. Type V (-5), bolt, self-locking, steel, 160 KSI Ftu, 250 °F, 12 point, external wrenching (externally wrenching cap screws).

(Use NASM21098 for material, heat treatment (if any), and sizes/lengths.)

A-A-60035 Part Number Zinc-Nickel Type K16 Plated (hex chrome-free)	NASM21098 Reference Dash Number Cadmium Plated (see paragraph 7.3)	A-A-60035 Part Number Zinc-Nickel Type K16 Plated (hex chrome-free)	NASM21098 Reference Dash Number Cadmium Plated (see paragraph 7.3)
AA60035-5-17QX	MS21098-17	AA60035-5-73QX	MS21098-73
AA60035-5-18QX	MS21098-18	AA60035-5-74QX	MS21098-74
AA60035-5-23QX	MS21098-23	AA60035-5-75QX	MS21098-75
AA60035-5-24QX	MS21098-24	AA60035-5-76QX	MS21098-76
AA60035-5-25QX	MS21098-25	AA60035-5-77QX	MS21098-77
AA60035-5-30QX	MS21098-30	AA60035-5-79QX	MS21098-79
AA60035-5-31QX	MS21098-31	AA60035-5-80QX	MS21098-80
AA60035-5-32QX	MS21098-32	AA60035-5-81QX	MS21098-81
AA60035-5-37QX	MS21098-37	AA60035-5-82QX	MS21098-82
AA60035-5-38QX	MS21098-38	AA60035-5-83QX	MS21098-83
AA60035-5-39QX	MS21098-39	AA60035-5-84QX	MS21098-84
AA60035-5-44QX	MS21098-44	AA60035-5-85QX	MS21098-85
AA60035-5-45QX	MS21098-45	AA60035-5-86QX	MS21098-86
AA60035-5-46QX	MS21098-46	AA60035-5-87QX	MS21098-87
AA60035-5-50QX	MS21098-50	AA60035-5-88QX	MS21098-88
AA60035-5-51QX	MS21098-51	AA60035-5-89QX	MS21098-89
AA60035-5-52QX	MS21098-52	AA60035-5-90QX	MS21098-90
AA60035-5-56QX	MS21098-56	AA60035-5-91QX	MS21098-91
AA60035-5-57QX	MS21098-57	AA60035-5-92QX	MS21098-92
AA60035-5-58QX	MS21098-58	AA60035-5-93QX	MS21098-93
AA60035-5-59QX	MS21098-59	AA60035-5-94QX	MS21098-94
AA60035-5-60QX	MS21098-60	AA60035-5-95QX	MS21098-95
AA60035-5-61QX	MS21098-61	AA60035-5-96QX	MS21098-96
AA60035-5-64QX	MS21098-64	AA60035-5-97QX	MS21098-97
AA60035-5-65QX	MS21098-65	AA60035-5-98QX	MS21098-98
AA60035-5-66QX	MS21098-66	AA60035-5-99QX	MS21098-99
AA60035-5-67QX	MS21098-67	AA60035-5-100QX	MS21098-100
AA60035-5-68QX	MS21098-68	AA60035-5-101QX	MS21098-101
AA60035-5-69QX	MS21098-69	AA60035-5-102QX	MS21098-102
AA60035-5-72QX	MS21098-72		

PIN construction:



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Examples of clean finish PIN's:

AA60035-5-83QX = Bolt, self-locking, 12 point, steel, .5000-20 UNF-3A thread, 1.750 inch long, zinc-nickel Type K16 plated.
AA60035-5-100QX = a NASM21098 – bolt, self-locking, 12 point, steel, 0.5000-20 UNF-3A thread, 3.000 inch long, zinc-nickel Type K16 plated.

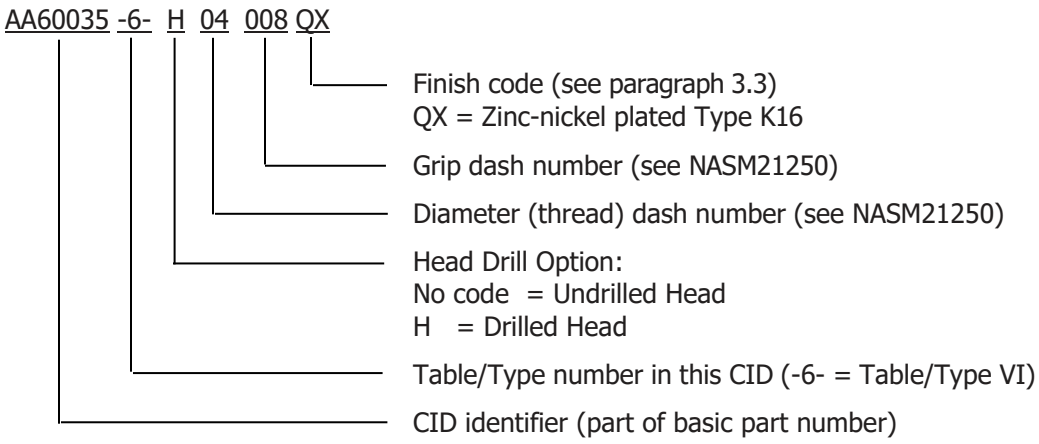
TABLE VI. Type VI (-6), bolt, tension, steel, external wrenching, flanged, 12-Point, 180 KSI Ftu, 450 ºF.

(Use NASM21250 for material, heat treatment (if any), and sizes/lengths.)

A-A-60035 Part Number Zinc-Nickel Type K16 Plated Undrilled Head (hex chrome-free)	NASM21250 Reference Dash Number Cadmium Plated (see paragraph 7.3) Undrilled Head	A-A-60035 Part Number Zinc-Nickel Type K16 Plated Drilled Head (hex chrome-free)	NASM21250 Reference Dash Number Cadmium Plated (see paragraph 7.3) Drilled Head
AA60035-6-02*QX	MS21250-02#	AA60035-6-H02*QX	MS21250H02#
AA60035-6-03*QX	MS21250-03#	AA60035-6-H03*QX	MS21250H03#
AA60035-6-04*QX	MS21250-04#	AA60035-6-H04*QX	MS21250H04#
AA60035-6-05*QX	MS21250-05#	AA60035-6-H05*QX	MS21250H05#
AA60035-6-06*QX	MS21250-06#	AA60035-6-H06*QX	MS21250H06#
AA60035-6-07*QX	MS21250-07#	AA60035-6-H07*QX	MS21250H07#
AA60035-6-08*QX	MS21250-08#	AA60035-6-H08*QX	MS21250H08#
AA60035-6-09*QX	MS21250-09#	AA60035-6-H09*QX	MS21250H09#
AA60035-6-10*QX	MS21250-10#	AA60035-6-H10*QX	MS21250H10#
AA60035-6-12*QX	MS21250-12#	AA60035-6-H12*QX	MS21250H12#
AA60035-6-14*QX	MS21250-14#	AA60035-6-H14*QX	MS21250H14#
AA60035-6-16*QX	MS21250-16#	AA60035-6-H16*QX	MS21250H16#
AA60035-6-18*QX	MS21250-18#	AA60035-6-H18*QX	MS21250H18#
AA60035-6-20*QX	MS21250-20#	AA60035-6-H20*QX	MS21250H20#
AA60035-6-22*QX	MS21250-22#	AA60035-6-H22*QX	MS21250H22#
AA60035-6-24*QX	MS21250-24#	AA60035-6-H24*QX	MS21250H24#

* See PIN construction below for part number breakdown
See NASM21250 for part number breakdown

PIN construction:



Examples of clean finish PIN's:

AA60035-6-02008QX = Bolt, undrilled head, .1640-32 UNJC-3A thread, .500 inch grip length, zinc-nickel Type K16 plated.
AA60035-6-H04008QX = Bolt, drilled head, .2500-28 UNJF-3A thread, .500 inch grip length, zinc-nickel Type K16 plated.

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TABLE VII. DELETED.

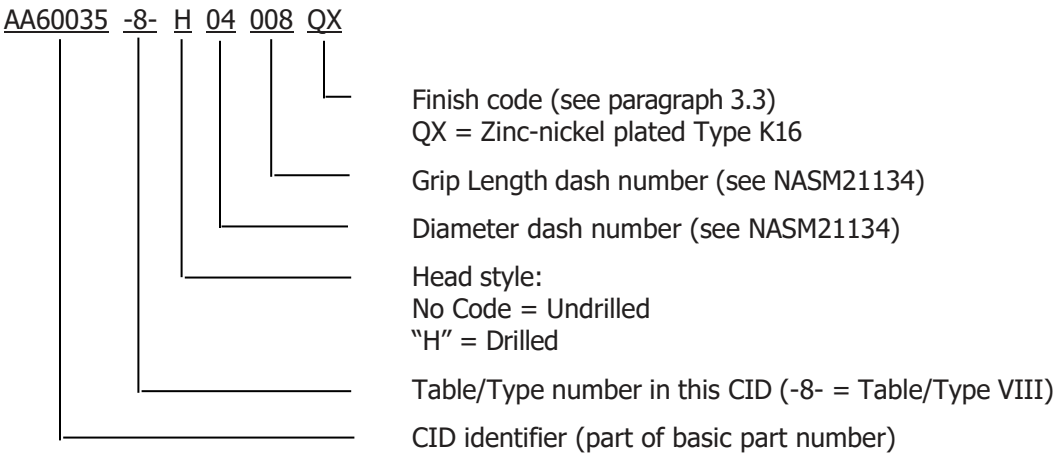
TABLE VIII. Type VIII (-8), bolt, tension, steel, 180 KSI Ftu, 450 °F external wrenching, spline drive, flanged.

(Use NASM21134 for material, heat treatment (if any), and sizes/lengths.)

A-A-60035 Part Number Zinc-Nickel Type K16 Plated Undrilled Head (hex chrome-free)	NASM21134 Reference Dash Number Cadmium Plated (see paragraph 7.3) Undrilled Head	A-A-60035 Part Number Zinc-Nickel Type K16 Plated Drilled Head (hex chrome-free)	NASM21134 Reference Dash Number Cadmium Plated (see paragraph 7.3) Drilled Head
AA60035-8-03*QX	MS21134-03#	AA60035-8-H03*QX	MS21134H03#
AA60035-8-04*QX	MS21134-04#	AA60035-8-H04*QX	MS21134H04#
AA60035-8-05*QX	MS21134-05#	AA60035-8-H05*QX	MS21134H05#
AA60035-8-06*QX	MS21134-06#	AA60035-8-H06*QX	MS21134H06#
AA60035-8-07*QX	MS21134-07#	AA60035-8-H07*QX	MS21134H07#
AA60035-8-08*QX	MS21134-08#	AA60035-8-H08*QX	MS21134H08#
AA60035-8-09*QX	MS21134-09#	AA60035-8-H09*QX	MS21134H09#
AA60035-8-10*QX	MS21134-10#	AA60035-8-H10*QX	MS21134H10#
AA60035-8-12*QX	MS21134-12#	AA60035-8-H12*QX	MS21134H12#
AA60035-8-14*QX	MS21134-14#	AA60035-8-H14*QX	MS21134H14#
AA60035-8-16*QX	MS21134-16#	AA60035-8-H16*QX	MS21134H16#
AA60035-8-18*QX	MS21134-18#	AA60035-8-H18*QX	MS21134H18#
AA60035-8-20*QX	MS21134-20#	AA60035-8-H20*QX	MS21134H20#
AA60035-8-22*QX	MS21134-22#	AA60035-8-H22*QX	MS21134H22#
AA60035-8-24*QX	MS21134-24#	AA60035-8-H24*QX	MS21134H24#

* See PIN Construction below for part number breakdown
\# See NASM21134 for part number breakdown

PIN construction:



Example of clean finish PIN's:

AA60035-8-04008QX = Bolt, .2500-28 UNJF-3A Thread, .970 long, .500 grip, undrilled head, zinc-nickel Type K16 plated.
AA60035-8H04008QX = Bolt, .2500-28 UNJF-3A, .970 long, .500 grip, drilled head, zinc-nickel Type K16 plated.

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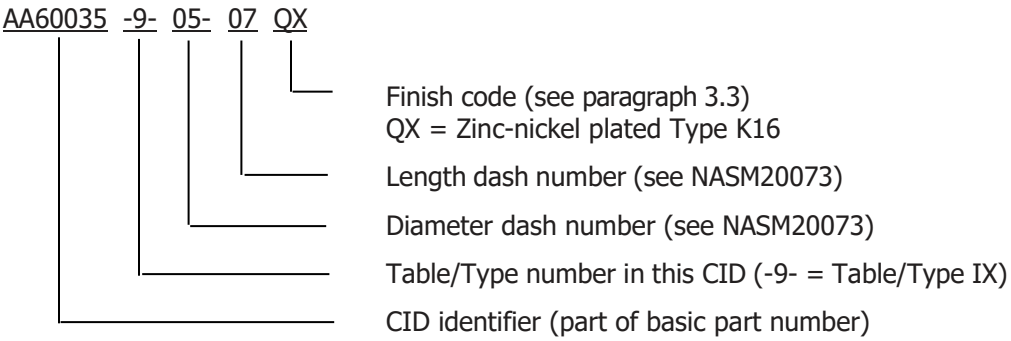
TABLE IX. Type IX (-9), bolt, machine, aircraft, drilled head, fine thread.

(Use NASM20073 for material, heat treatment (if any), and sizes/lengths.)

A-A-60035 Part Number Zinc-Nickel Type K16 Plated (hex chrome-free)	NASM20073 Reference Dash Number Cadmium Plated (see paragraph 7.3)
Steel	
AA60035-9-03-*QX	MS20073-03-*
AA60035-9-04-*QX	MS20073-04-*
AA60035-9-05-*QX	MS20073-05-*
AA60035-9-06-*QX	MS20073-06-*
AA60035-9-07-*QX	MS20073-07-*
AA60035-9-08-*QX	MS20073-08-*
AA60035-9-09-*QX	MS20073-09-*
AA60035-9-10-*QX	MS20073-10-*
AA60035-9-12-*QX	MS20073-12-*

* Indicates length dash number (numerous possibilities; refer to NASM20073)

PIN construction:



Example of clean finish PIN's:

AA60035-9-05-07QX = Bolt, .3125-24 UNF-3A thread, .375 grip, .922 long, zinc-nickel Type K16 plated.

AA60035-9-08-14QX = Bolt, .5000-20 UNF-3A thread, .750 grip, 1.547 long, zinc-nickel Type K16 plated.

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7.2 Source of documents.

7.2.1 FAR: Copies of the Federal Acquisition Regulations are available online from https://www.acquisition.gov/.

7.2.2 Military specifications: Copies of military specifications are available online from https://quicksearch.dla.mil.

7.2.3 AIA documents: Copies of Aerospace Industries Association documents are available online from https://www.aia-aerospace.org/standards/.

7.3 Hazardous materials. Cadmium or zinc plating with hexavalent chromium is not recommended for new design on Army ground systems.

7.4 Ordering data. The contract or order should specify the following:
a. Title, number, and date of this CID.
b. National stock number (if available).
c. Material.
d. Hexavalent chromium and cadmium free part number (PIN) from this CID.
e. Quantity.

7.5 Key words:
Alloy steel
Carbon steel
Plating
Steel
Zinc-Nickel

7.6 Changes from previous issue. The margins of this CID are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

MILITARY INTERESTS:

Custodians:
Army – AT
Navy – AS
Air Force – 71
DLA – CC

Review activities:
Army – AR, AV, CR, MI
Navy – CG, MC, YD
Air Force – 03, 11, 70
DLA – GS4, GS7

CIVIL AGENCY COORDINATING ACTIVITY:
GSA-FAS

Preparing Activity:
Army – AT

(Project 5306-2026-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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